

Version 2 synthetic mortgage-lending exploratory analysis

This notebook connects the main Version 2 research layers:

1. synthetic applicants and fixed loan requests;
2. hidden conditional repayment risk and realized repayment outcomes;
3. traditional versus flexible-ML risk estimates;
4. budget-constrained portfolio allocation;
5. direct belief distortion and upstream opportunity mechanisms; and
6. the corrected five-replication systemic checkpoint.

The purpose is descriptive and diagnostic. The notebook does not train models, regenerate data, rerun optimization, or modify source artifacts.

Interpretation boundaries

- Groups **A** and **B** are researcher-defined synthetic groups, not empirical racial groups.
- **rho** is the probability of making the next scheduled payment conditional on having made all previous payments.
- The oracle uses hidden simulation truth and is an evaluator benchmark, not a feasible lender.
- Direct and systemic mechanisms are present only because the simulation explicitly encodes them.
- The repeated-seed section contains five corrected replications. It is an interim checkpoint, not the planned final 50-replication analysis.

1. Setup and source inventory

The loader finds the repository root whether the notebook is opened from the root directory or from `notebooks/`. All inputs are read-only CSV artifacts already produced by the project pipeline.

	dataset	rows	columns	memory_mb	relative_path
0	applicants	10000	11	2.6581	data/synthetic/economic_lending/v2_baseline/applicants.csv
1	loans	10000	10	2.9801	data/synthetic/economic_lending/v2_baseline/loan_options.csv
2	truth	10000	3	0.8701	data/synthetic/economic_lending/v2_baseline/simulation_truth.csv
3	outcomes	10000	5	0.9601	data/synthetic/economic_lending/v2_baseline/loan_outcomes.csv
4	risk_additive	6	19	0.0018	results/tables/v2_risk_model_comparison.csv
5	risk_nonlinear	6	15	0.0016	results/tables/v2_nonlinear_risk_model_comparison.csv
6	portfolio	18	27	0.0087	results/tables/v2_portfolio_summary.csv
7	regret	18	8	0.0046	results/tables/v2_portfolio_regret.csv
8	direct	72	20	0.0254	results/tables/v2_direct_group_effects.csv
9	systemic_access	8	6	0.0009	results/tables/v2_systemic_opportunity_summary.csv
10	systemic_switchers	16	35	0.0067	results/tables/v2_systemic_switcher_effects.csv
11	systemic_checkpoint	488	13	0.2213	results/tables/v2_systemic_mc_corrected_checkpoint_summary.csv

2. Integrity checks and analysis table

The four borrower-level files should have one row per applicant and join one-to-one. The checks also verify the repayment identity

$P(\text{complete all payments}) = \rho^T$

and confirm that completed loans have no recorded default period.

Merged analysis table: 10,000 applicants × 28 columns

check		passed
0	10,000 applicants	True
1	applicant IDs unique	True
2	loan IDs unique	True
3	truth IDs unique	True
4	outcome IDs unique	True
5	all applicant key sets match	True
6	ρ lies in [0, 1]	True
7	full-repayment probability lies in [0, 1]	True
8	loan principal is positive	True
9	completed loans have no default period	True
10	full repayment equals ρ ** term	True

showing first 12 of 28 columns

	applicant_id	group	cohort	age_years	annual_income	credit_score	employment_years	liquid_assets	existing_monthly_debt	property_value	requested_loan_amount	option_id
0	V2APP00000001	B	historical_validation	66	68,965.8900	724	28.7350	17,248.8200	1,486.3100	304,310.1900	255,305.4100	V2APP0000000001:requested
1	V2APP00000002	A	historical_train	56	134,412.5300	696	30.4090	99,053.5200	1,000.4500	502,479.4700	335,124.8600	V2APP0000000002:requested
2	V2APP00000003	A	historical_train	57	93,670.7200	757	12.4070	58,622.4500	804.8500	361,148.3900	226,824.6600	V2APP0000000003:requested
3	V2APP00000004	B	historical_train	32	80,245.1900	676	8.0070	22,684.8400	704.5300	332,480.6700	289,663.8400	V2APP0000000004:requested
4	V2APP00000005	A	historical_train	70	296,967.2100	759	52.0000	122,347.9500	2,608.2100	871,630.7300	561,936.3700	V2APP0000000005:requested

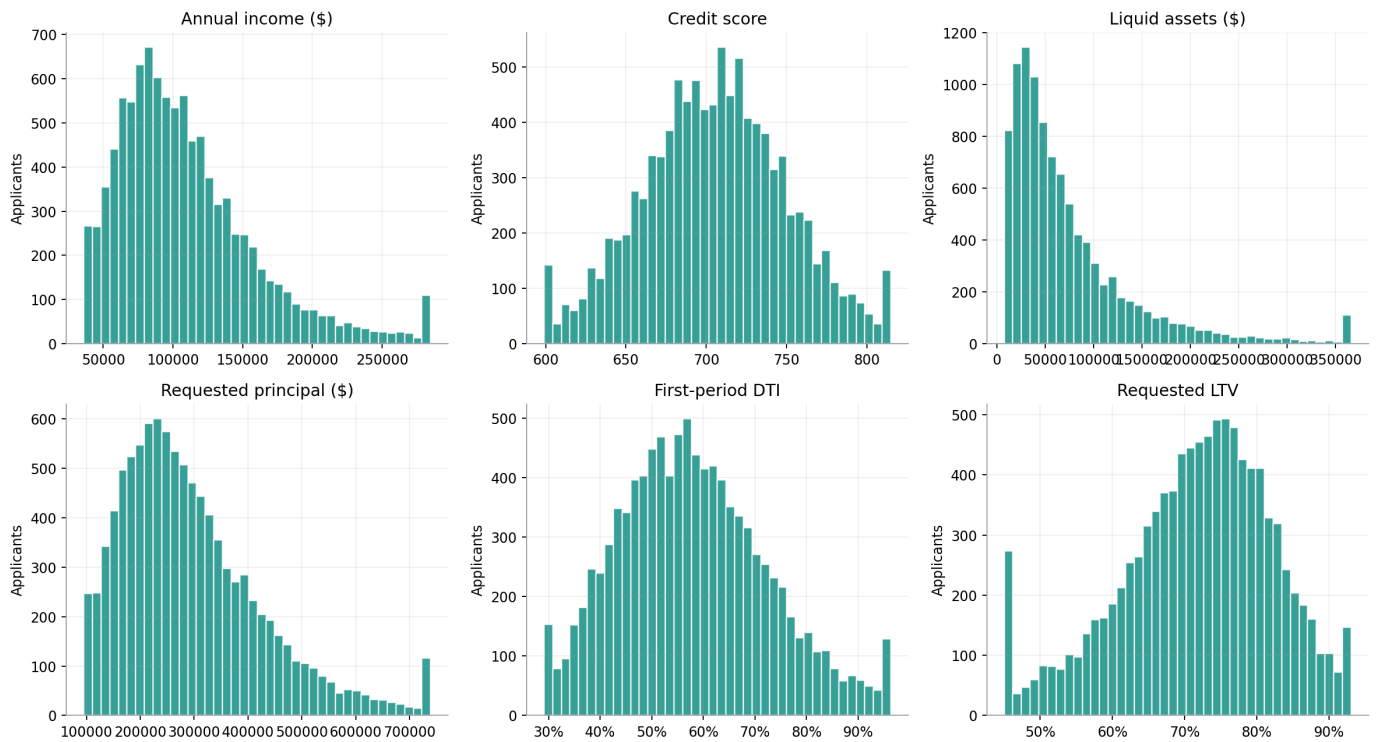
3. Population composition and financial distributions

The cohort split occurs at the applicant level, which prevents payment rows from the same borrower from crossing training, validation, and evaluation samples.

group	A	B	All
cohort			
evaluation	986	1014	2000
historical_train	3001	2999	6000
historical_validation	1046	954	2000
All	5033	4967	10000

	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max
annual_income	10,000.0000	109,238.9808	49,871.5338	25,000.0000	36,030.6163	47,694.0720	73,941.2250	99,459.3200	133,478.3725	207,490.3980	284,851.8934	300,000.0000
credit_score	10,000.0000	705.4999	45.6405	530.0000	599.0000	630.0000	675.0000	706.0000	735.0000	781.0000	815.0000	850.0000
employment_years	10,000.0000	18.1059	8.9155	0.3200	2.8568	5.4240	11.3648	17.0660	23.8470	34.2231	42.3730	53.0000
liquid_assets	10,000.0000	74,842.8781	71,792.8503	1,754.1300	7,633.2675	13,360.5080	30,033.1500	53,432.9000	93,546.7425	206,704.3305	366,185.1131	750,000.0000
existing_monthly_debt	10,000.0000	1,386.2452	678.2373	161.2200	406.2665	556.1110	904.1200	1,247.0950	1,726.1900	2,680.2220	3,589.0073	6,832.5100
property_value	10,000.0000	419,452.1396	200,458.0825	100,000.0000	130,972.9175	175,809.3275	277,879.2950	379,488.2400	512,976.1400	810,818.7485	1,089,468.3748	1,500,000.0000
requested_principal	10,000.0000	295,119.6724	135,545.4692	48,164.8900	94,962.8237	127,429.3905	198,207.1550	268,262.9400	363,808.9475	550,400.8185	739,136.4290	1,124,751.8800
first_period_dti	10,000.0000	0.5810	0.1475	0.2042	0.2912	0.3606	0.4751	0.5700	0.6739	0.8424	0.9636	1.2112
requested_ltv	10,000.0000	0.7168	0.1066	0.4500	0.4500	0.5101	0.6524	0.7274	0.7917	0.8768	0.9308	0.9500

Baseline borrower and loan distributions (1st-99th percentile display range)



Baseline group-balance diagnostic

For each continuous variable, the standardized mean difference is

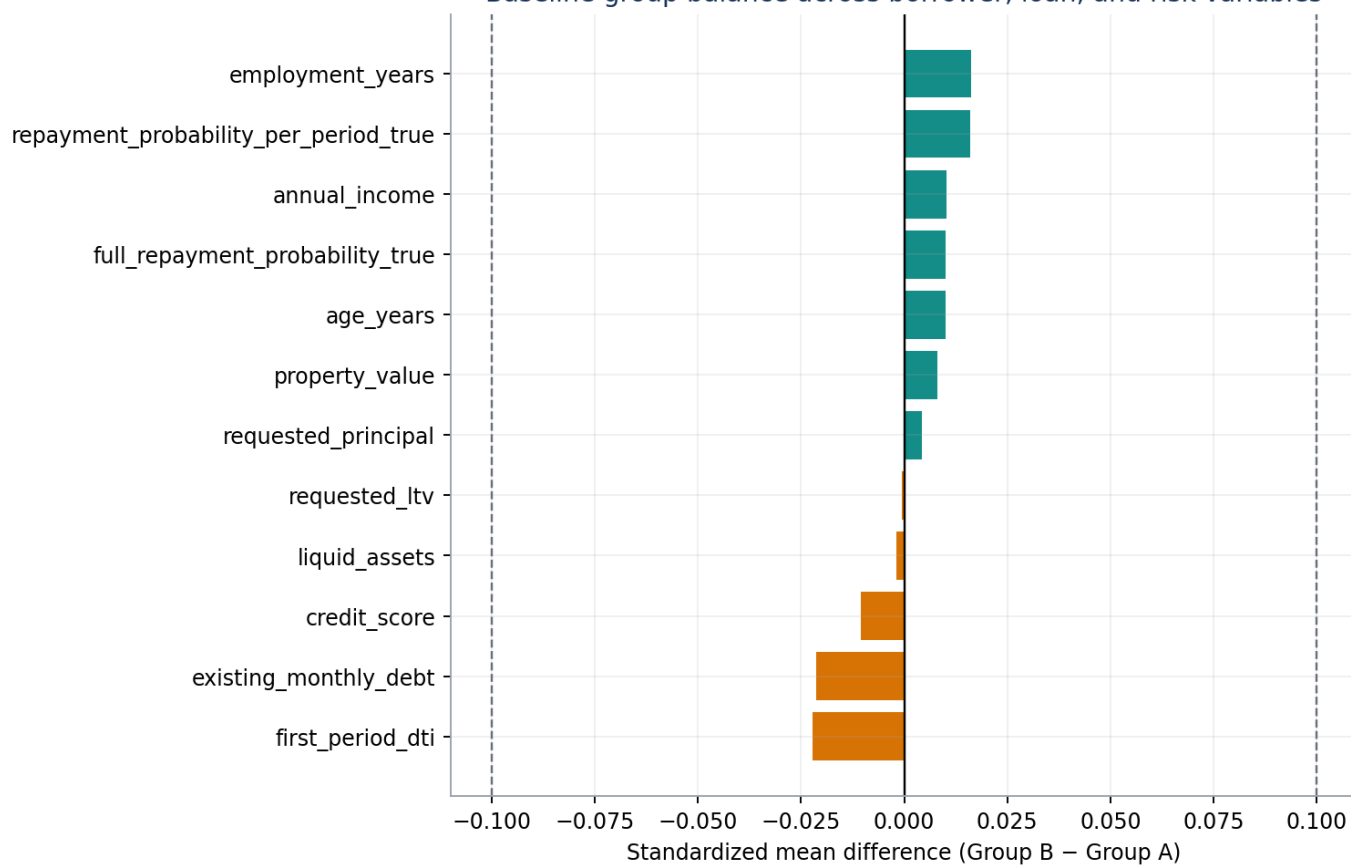
$$\text{Standardized mean difference} = (\text{mean}_B - \text{mean}_A) / \text{pooled SD}$$

This is a descriptive balance check. It is not a test of real-world group equality.

Largest absolute baseline SMD: 0.022

	variable	group_a_mean	group_b_mean	smd_b_minus_a	absolute_smd
8	first_period_dti	0.5826	0.5794	-0.0223	0.0223
5	existing_monthly_debt	1,393.4276	1,378.9674	-0.0213	0.0213
3	employment_years	18.0339	18.1789	0.0163	0.0163
10	repayment_probability_per_period_true	0.9926	0.9927	0.0160	0.0160
2	credit_score	705.7359	705.2607	-0.0104	0.0104
1	annual_income	108,984.1220	109,497.2261	0.0103	0.0103
11	full_repayment_probability_true	0.5343	0.5369	0.0101	0.0101
0	age_years	49.0872	49.2013	0.0100	0.0100
6	property_value	418,654.5468	420,260.3306	0.0080	0.0080
7	requested_principal	294,831.1279	295,412.0510	0.0043	0.0043
4	liquid_assets	74,906.9310	74,777.9741	-0.0018	0.0018
9	requested_ltv	0.7168	0.7167	-0.0006	0.0006

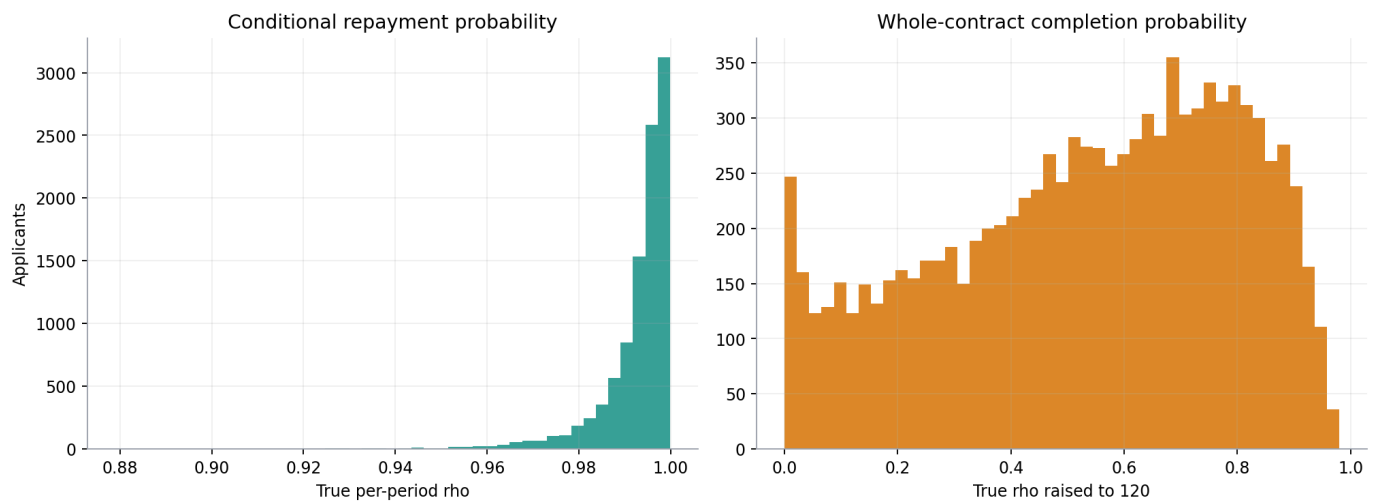
Baseline group balance across borrower, loan, and risk variables



4. Hidden repayment risk and realized outcomes

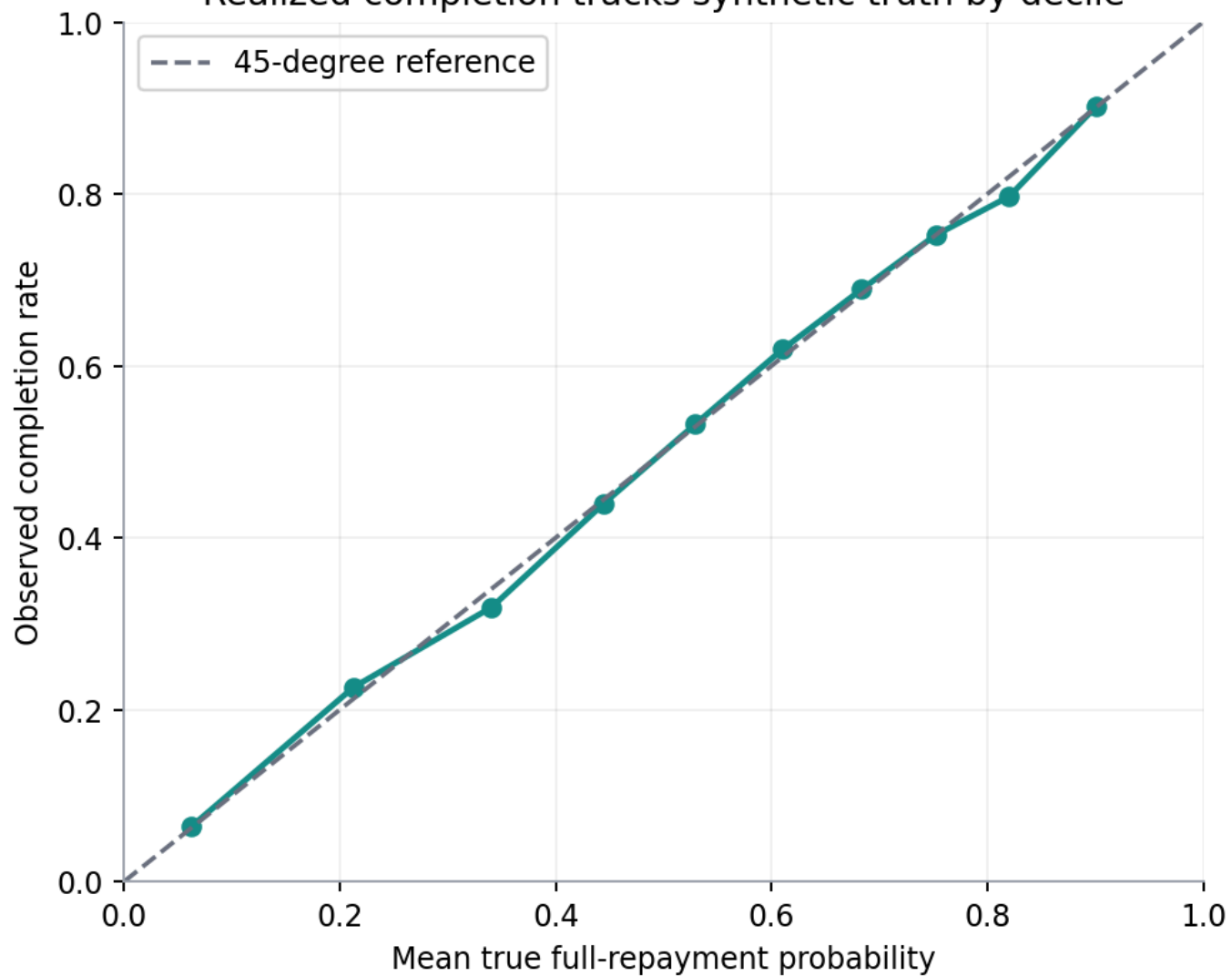
The true per-period probability is close to one for most applicants, but a 120-period horizon compounds small differences. Consequently, whole-loan completion probabilities are much more dispersed than per-period probabilities.

	metric	value
0	Mean conditional repayment probability	0.9926
1	Mean full-repayment probability	0.5356
2	Observed full-completion rate	0.5340
3	Mean periods paid	86.2970
4	Mean realized profit	23,595.8414
5	Share with positive realized profit	0.6296



	n	mean_true_full_repayment	observed_completion_rate
0	1000	0.0624	0.0640
1	1000	0.2126	0.2260
2	1000	0.3405	0.3190
3	1000	0.4447	0.4400
4	1000	0.5292	0.5320
5	1000	0.6100	0.6190
6	1000	0.6834	0.6890
7	1000	0.7520	0.7520
8	1000	0.8202	0.7970
9	1000	0.9009	0.9020

Realized completion tracks synthetic truth by decile



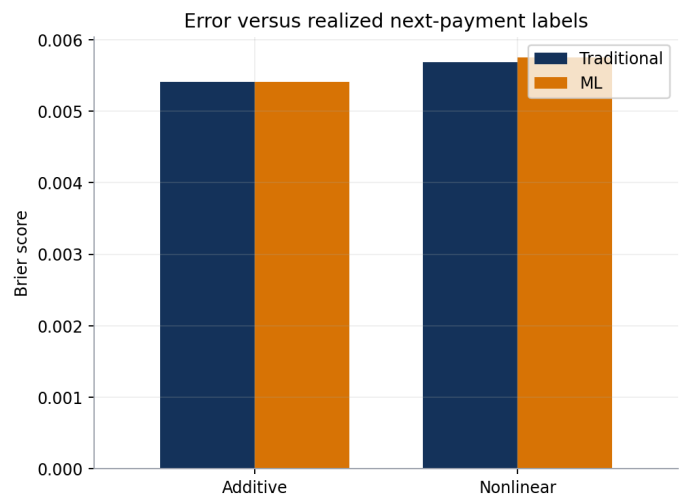
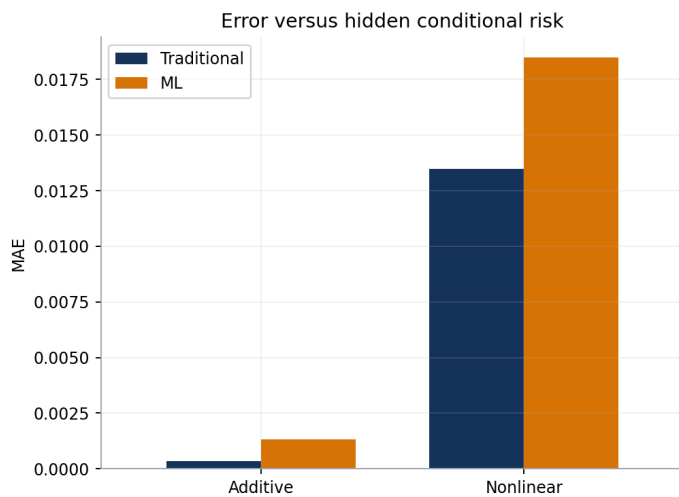
5. Risk-model recovery

Two evaluations are intentionally separate:

- probability recovery compares lender estimates with hidden `rho_true` ;
- realized-label metrics compare probabilities with stochastic next-payment outcomes.

Even the oracle cannot perfectly predict Bernoulli outcomes.

	world	model_label	mae	rmse	brier_score	log_loss	roc_auc
0	Additive	Traditional	0.0004	0.0007	0.0054	0.0318	0.7423
1	Additive	ML	0.0013	0.0030	0.0054	0.0318	0.7405
2	Additive	Oracle			0.0054	0.0318	0.7424
3	Nonlinear	ML	0.0185	0.0741	0.0057	0.0305	0.8309
4	Nonlinear	Oracle			0.0056	0.0301	0.8351
5	Nonlinear	Traditional	0.0135	0.0513	0.0057	0.0305	0.8336



6. Prediction error and portfolio value

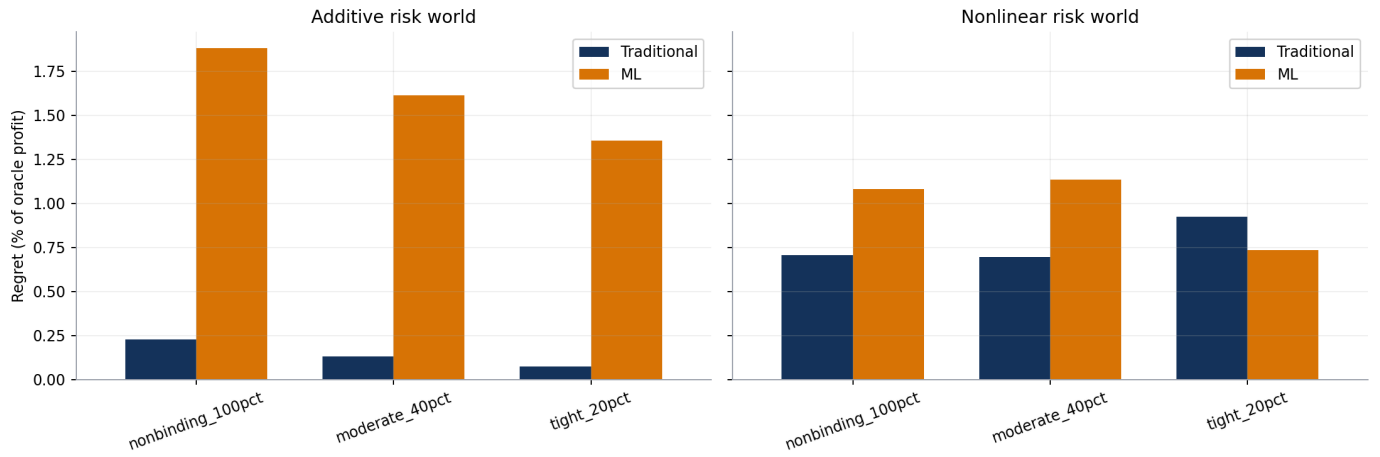
The portfolio optimizer maximizes perceived expected profit subject to the lending budget. Regret evaluates the selected portfolio using hidden truth:

Regret = oracle true expected profit - policy true expected profit.

Lower regret is better. The oracle has zero regret by construction.

	world_id	budget_id	policy	applicants_funded	funding_rate	budget_utilization	true_expected_portfolio_profit	truly_nonpositive_funded_fraction
0	additive_logistic_baseline	nonbinding_100pct	oracle	1281	0.6405	0.6703	91,464,741.0144	0.0000
1	additive_logistic_baseline	nonbinding_100pct	traditional	1295	0.6475	0.6825	91,256,425.7982	0.0224
2	additive_logistic_baseline	nonbinding_100pct	ml	1286	0.6430	0.6783	89,744,792.2289	0.0498
3	additive_logistic_baseline	moderate_40pct	oracle	736	0.3680	1.0000	73,957,837.7803	0.0000
4	additive_logistic_baseline	moderate_40pct	traditional	721	0.3605	1.0000	73,860,112.4352	0.0000
5	additive_logistic_baseline	moderate_40pct	ml	736	0.3680	1.0000	72,765,947.5446	0.0014
6	additive_logistic_baseline	tight_20pct	oracle	356	0.1780	1.0000	43,561,524.3882	0.0000
7	additive_logistic_baseline	tight_20pct	traditional	344	0.1720	1.0000	43,528,843.2856	0.0000
8	additive_logistic_baseline	tight_20pct	ml	373	0.1865	1.0000	42,970,348.8129	0.0000
9	nonlinear_v1	nonbinding_100pct	oracle	1237	0.6185	0.6270	108,247,282.1667	0.0000
10	nonlinear_v1	nonbinding_100pct	traditional	1181	0.5905	0.5979	107,482,461.9054	0.0042
11	nonlinear_v1	nonbinding_100pct	ml	1205	0.6025	0.6117	107,078,199.5840	0.0174
12	nonlinear_v1	moderate_40pct	oracle	758	0.3790	1.0000	86,639,282.3060	0.0000
13	nonlinear_v1	moderate_40pct	traditional	787	0.3935	1.0000	86,036,508.5106	0.0000
14	nonlinear_v1	moderate_40pct	ml	761	0.3805	1.0000	85,656,571.8267	0.0000
15	nonlinear_v1	tight_20pct	oracle	369	0.1845	1.0000	48,689,269.8028	0.0000
16	nonlinear_v1	tight_20pct	traditional	396	0.1980	1.0000	48,238,912.8750	0.0000
17	nonlinear_v1	tight_20pct	ml	364	0.1820	1.0000	48,332,163.1001	0.0000

Economic regret depends on the risk world and capital constraint



	world_id	budget_id	lower_regret_policy	regret_dollars	regret_percent_oracle
0	additive_logistic_baseline	moderate_40pct	traditional	97,725.3451	0.1321
1	additive_logistic_baseline	nonbinding_100pct	traditional	208,315.2162	0.2278
2	additive_logistic_baseline	tight_20pct	traditional	32,681.1026	0.0750
3	nonlinear_v1	moderate_40pct	traditional	602,773.7954	0.6957
4	nonlinear_v1	nonbinding_100pct	traditional	764,820.2612	0.7065
5	nonlinear_v1	tight_20pct	ml	357,106.7027	0.7334

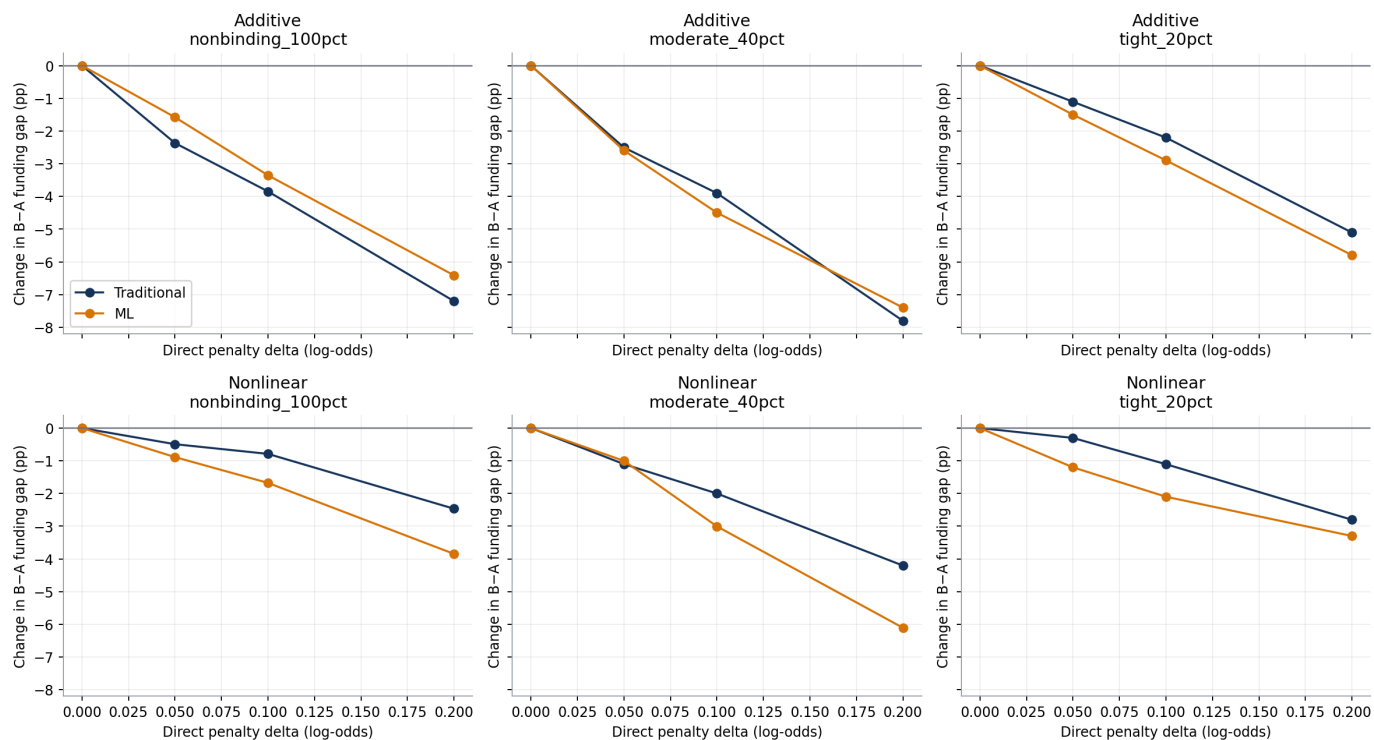
7. Direct belief distortion

The direct experiment reduces Group B's perceived repayment probability on the log-odds scale while leaving true borrower risk unchanged:

$$\text{logit}(\rho_{\text{used}}) = \text{logit}(\rho_{\text{base}}) - \delta \times I(\text{Group B}).$$

The plots show the change in the Group B-minus-Group A funding gap relative to the same policy at $\delta = 0$.

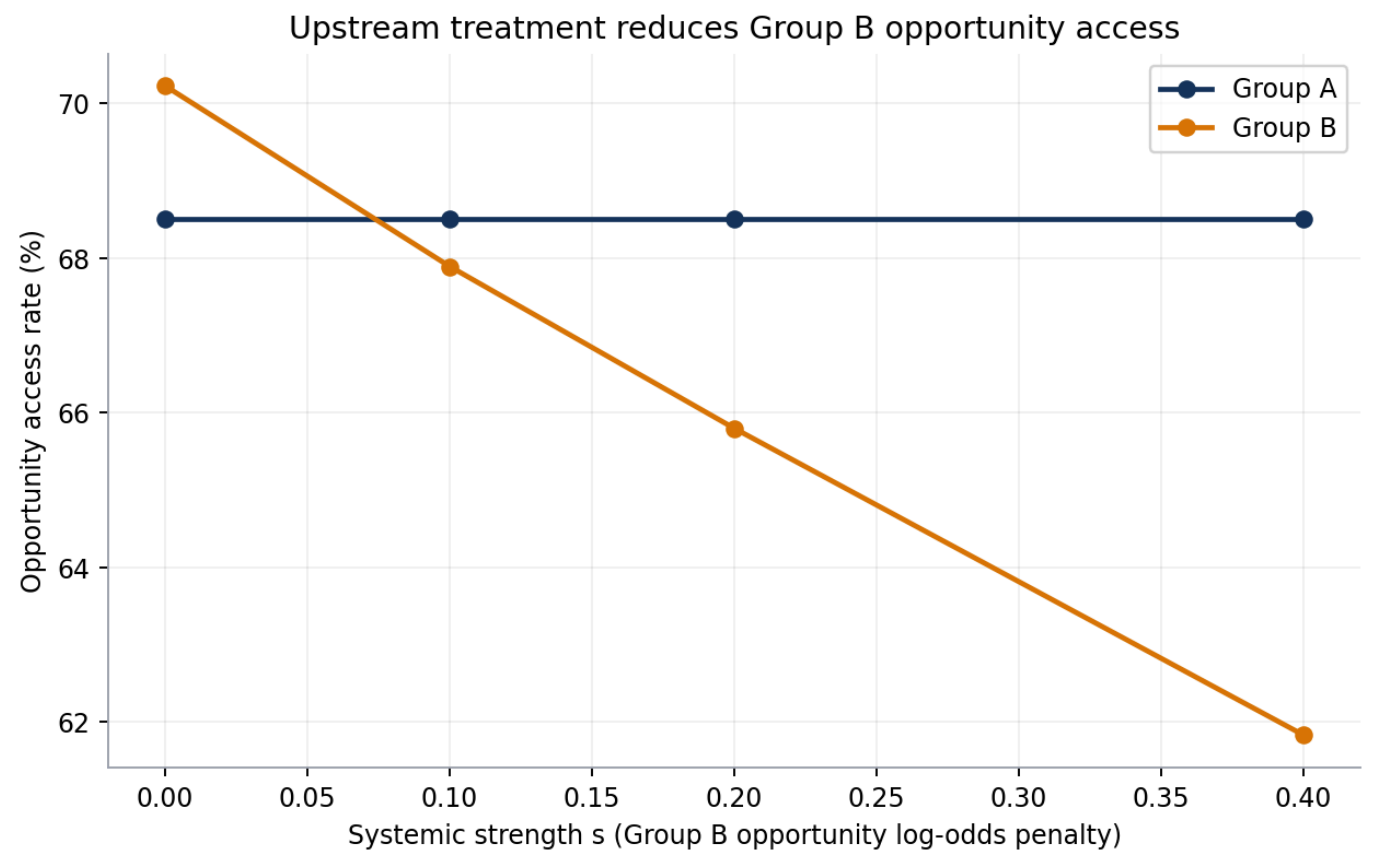
Direct belief distortion shifts funding away from Group B



	world_id	family	budget_id	group_a_change_pp	group_b_change_pp	gap_change_pp
3	additive_logistic_baseline	traditional	nonbinding_100pct	0.0000	-7.1992	-7.1992
7	additive_logistic_baseline	traditional	moderate_40pct	3.9554	-3.8462	-7.8015
11	additive_logistic_baseline	traditional	tight_20pct	2.6369	-2.4655	-5.1024
15	additive_logistic_baseline	ml	nonbinding_100pct	0.0000	-6.4103	-6.4103
19	additive_logistic_baseline	ml	moderate_40pct	3.5497	-3.8462	-7.3958
23	additive_logistic_baseline	ml	tight_20pct	2.6369	-3.1558	-5.7927
39	nonlinear_v1	traditional	nonbinding_100pct	0.0000	-2.4655	-2.4655
43	nonlinear_v1	traditional	moderate_40pct	2.2312	-1.9724	-4.2036
47	nonlinear_v1	traditional	tight_20pct	1.5213	-1.2821	-2.8033
51	nonlinear_v1	ml	nonbinding_100pct	0.0000	-3.8462	-3.8462
55	nonlinear_v1	ml	moderate_40pct	3.2454	-2.8600	-6.1054
59	nonlinear_v1	ml	tight_20pct	1.6227	-1.6765	-3.2992

8. Upstream opportunity mechanism

The systemic experiment acts earlier in the causal chain. It lowers Group B's probability of receiving a favorable opportunity state, which then changes financial characteristics, true repayment risk, expected profit, and potentially funding. There is no direct Group B term in the lender's final repayment decision in this mechanism.

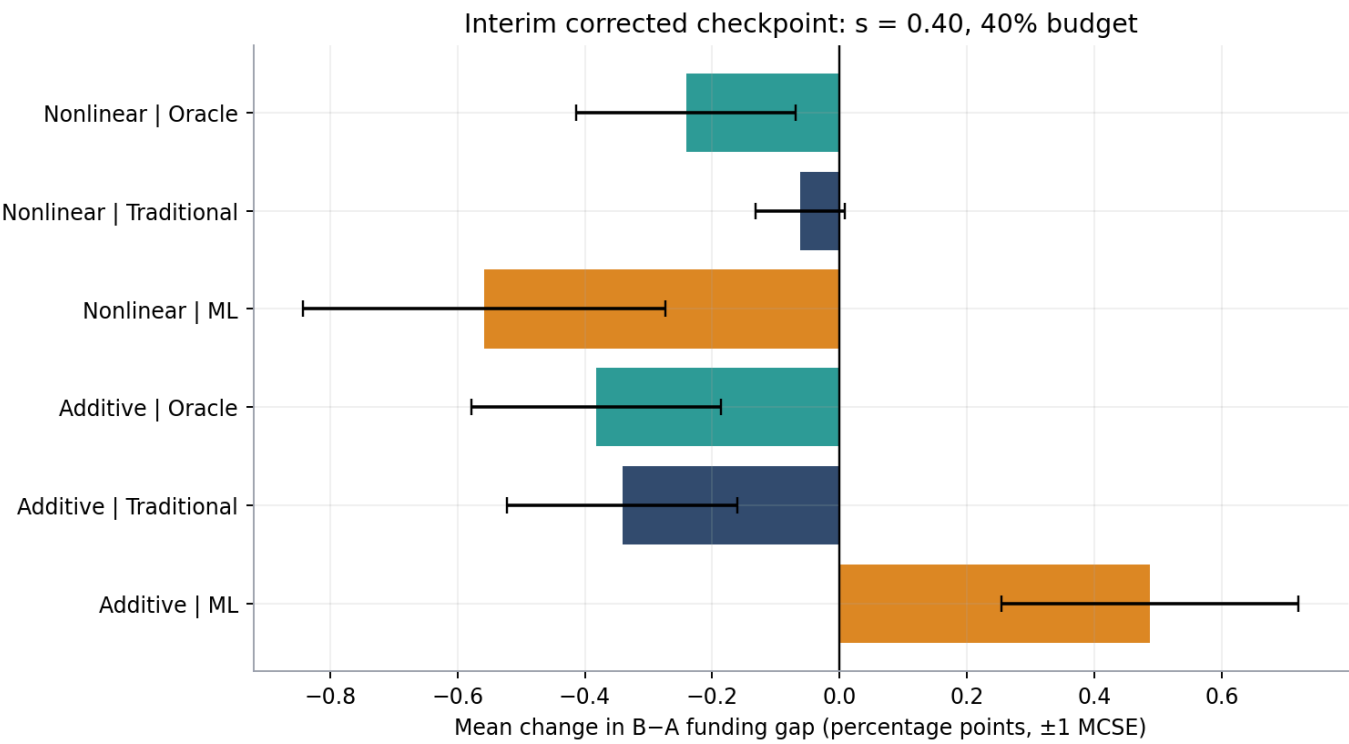


risk_world	additive_logistic_baseline	nonlinear_v1
metric		
n	417.0000	417.0000
share_of_group_b	0.0840	0.0840
mean_change_employment_years	-1.4807	-1.4807
mean_change_annual_income	-8,294.9222	-8,294.9222
mean_change_liquid_assets	-9,676.9650	-9,676.9650
mean_change_property_value	-23,778.7101	-23,778.7101
mean_change_requested_loan_amount	-13,626.9075	-13,626.9075
mean_change_first_period_dti	0.0125	0.0125
mean_change_requested_ltv	0.0080	0.0080
mean_change_full_repayment_probability_true	-0.0249	-0.0327
mean_change_expected_profit_true	-8,184.7175	-11,349.2535

9. Corrected repeated-seed systemic checkpoint

This section summarizes five validated corrected replications. Error bars show Monte Carlo standard errors, not confidence intervals about a real population. With only five replications, signs can remain unstable and model rankings should not be generalized.

	label	n_corrected	mean_pp	mcse_pp	minimum_pp	maximum_pp
243	Additive ML	5	0.4873	0.2334	-0.1079	0.9862
246	Additive Traditional	5	-0.3413	0.1812	-1.0041	0.0991
249	Additive Oracle	5	-0.3825	0.1962	-1.1031	0.0986
279	Nonlinear ML	5	-0.5582	0.2841	-1.1974	0.5029
282	Nonlinear Traditional	5	-0.0614	0.0699	-0.3112	0.1014
285	Nonlinear Oracle	5	-0.2409	0.1726	-0.7045	0.1980



10. Overall findings from the current synthetic snapshot

The following cell derives a concise set of findings directly from the loaded files, so the numbers update when the notebook is rerun against compatible artifacts.

1. Baseline balance is strong by construction: the largest absolute Group B-minus-A standardized mean difference is 0.022.
2. The mean true whole-loan completion probability is 53.6%; the realized completion rate is 53.4%. The small difference is compatible with Bernoulli sampling variation.
3. In the additive world, traditional-logit risk MAE is 0.000364, versus 0.001341 for ML.
4. In the nonlinear world, traditional risk MAE is 0.013471, versus 0.018465 for ML. Flexibility does not guarantee better recovery under the current sample and tuning design.
5. Traditional has lower portfolio regret in 5 of the 6 world-budget cells; ML has lower regret in 1. The exception shows that model rankings can depend on the economic decision environment.
6. At direct penalty $\delta=0.20$ in the additive traditional 40% budget cell, Group A funding changes by +3.96 pp, Group B by -3.85 pp, and the B-minus-A gap changes by -7.80 pp.
7. At systemic strength 0.40, Group B opportunity access falls from 70.2% to 61.8%. Among 417 switchers, mean income changes by \$-8,295, liquid assets by \$-9,677, and employment by -1.48 years.
8. The corrected repeated-seed checkpoint has five replications. It is useful for identifying instability, but it is not sufficient for final uncertainty or model-ranking claims.

11. Research interpretation

The current results support three distinctions:

1. **Risk prediction and economic value are related but not identical.** A model with lower probability error usually has an advantage, but the portfolio objective depends on ranking, loan size, expected profit, and the budget constraint.
2. **Direct and upstream mechanisms answer different questions.** Direct belief distortion changes the lender's treatment of otherwise comparable applicants. The systemic mechanism changes the economic conditions applicants bring into underwriting.
3. **Single-population findings are not uncertainty estimates.** The corrected five-replication checkpoint begins to show seed-to-seed variation, but the predeclared 50-replication study remains incomplete.

These are properties of the researcher-defined synthetic environment. They do not establish discrimination in real mortgage markets, and adjusted or unadjusted differences should not automatically be interpreted as causal estimates.